higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING
NQF LEVEL 2

(12050652)

13 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only a BLACK or BLUE ink.
6. Write neatly and legibly.

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QUESTION 1: INTRODUCTION TO CIVIL AND STRUCTURAL STEEL WORK

1.1 Make a neat freehand sketch of each of the following:

1.1.1 H-channel

1.1.2 I-section

1.1.3 T-section

1.1.4 Equal-angle leg

1.1.5 Unequal-angle leg

(5 × 2) (10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 The term *structural steel* refers to the various products of a steel mill such as sections, bars and plates.

1.2.2 Steel is not a human-made metal derived from iron.

1.2.3 Channels can be compared to an I-beam that has been timed on one side to give a flat-back web.

1.2.4 The two main classes of profiles are hot-rolled sections and cold-formed sections.

1.2.5 Structural steel is produced in a number of different strength grades.

(5 × 1) (5)

1.3 Use a diagram or flow chart to outline the duties of FOUR role players involved in a construction project. (8)

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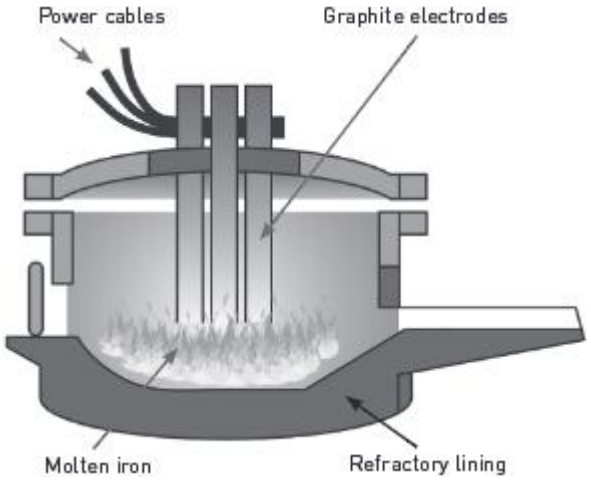
1.4 Choose a term from COLUMN B that matches an abbreviation in COLUMN A. Write only the letter (A–L) next to the question number (1.4.1–1.4.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.4.1	ms.	A	column
1.4.2	NF	B	mild steel
1.4.3	COL	C	brickwork
1.4.4	bwk	D	millimetre
1.4.5	RE	E	cleat
1.4.6	mm	F	not to scale
1.4.7	AB	G	aluminium
1.4.8	rc.	H	rodding eye
1.4.9	al.	I	arrangement
1.4.10	NTS	J	reinforced concrete
		K	airbrick
		L	near face

(10 × 1) (10)

1.5 Name each of the following:

1.5.1



1.5.2



(2 × 1) (2)

1.6 Name FIVE types of corrosion-/decay-prevention methods in concrete. (5)
[40]

QUESTION 2: STRUCTURAL STEEL WORK AND REINFORCED CONCRETE

2.1 Explain the purpose of reinforcing concrete. (2)

2.2 Briefly explain each of the following symbols:

2.2.1 Y 32

2.2.2 R 16

(2 × 1) (2)

2.3 Make a comparison between *hot-rolled* and *cold-formed* steel. (2 + 2) (4)

2.4 Define *workability* with reference to concrete. (2)

2.5 Steel profiles are widely used in the construction of railways, roads and other infrastructures, appliances and buildings.

Name FIVE other common applications of steel. (5)

2.6 Name an instrument used to measure the strain or lengthening of steel. (1)

2.7 Indicate whether the following statement is TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.7) in the ANSWER BOOK.

Recycling steel repeatedly does not reduce its strength. (1)

2.8 Briefly explain the purpose of conducting cube tests on concrete. (2)

2.9 Sketch THREE different types of slumps in a slump test. (3 × 3) (9)

2.10 Name TWO tests that can be done on concrete and give a reason why each is done. (2 × 2) (4)

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2.11 Give THREE reasons why concrete should be used as a construction material. (3)

[35]

QUESTION 3: BOLTS IN STRUCTURAL STEEL WORK

Make a neat drawing of a typical structural bolt, nut and washer assembly. **[10]**

QUESTION 4: WELDING SYMBOLS AND JOINTS

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.1–4.2) in the ANSWER BOOK.

4.1 Welding symbols are developed by the Institute of Welding.

4.2 The reference line of a welding symbol identifies the side of the joint to be welded.

(2 × 1) (2)

4.3 Choose a picture from COLUMN B that matches a description in COLUMN A. Write only the letter (A–E) next to the question number (4.3.1–4.3.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
4.3.1	Fillet	A	
4.3.2	Bevel		
4.3.3	Convex	B	
4.3.4	Weld all round		
4.3.5	Vee	C	
		D	
		E	

(5 × 1) (5)

4.4 Name THREE types of basic surface contours. (3)

4.5 Name FIVE basic elements of a welding symbol. (5)

[15]

TOTAL: 100

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